DESCRIPTIVE REPORT

RetailPulse Sales Analysis

Descriptive Report using IBM Cognos Analytics and Power BI

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Course: Business Analytics

Submission Date: 13 October 2026

1. Introduction and Objective

This report presents the descriptive analysis completed for the RetailPulse Sales dataset using IBM Cognos Analytics and Microsoft Power BI. The work covers five Cognos reporting practicals and a Power BI analysis built from the available exported RetailPulse report data. The report focuses on sales by location and product category and documents both successful outputs and Cognos execution issues encountered during the practicals.

Objectives

- Analyse the available RetailPulse sales information using business analytics tools.
- Create and document list, grouped, filtered, crosstab and prompt reports in IBM Cognos Analytics.
- Compare sales performance across the available locations and product categories.
- Create Power BI visuals and a DAX measure for overall sales.
- Identify descriptive findings and translate them into practical recommendations.
- Document technical limitations without inventing results that could not be verified.

Report scope

The report documents all five practicals. Where a practical could not be executed successfully because Cognos returned a server error, the report records the error as a technical limitation rather than presenting an unverified result.

Practical completion summary

Practical	Report	Evidence / status
1	FirstSalesReport	Successful output PDF
2	SecondSalesReport	Server error on execution
3	ThirdSalesReport	Server error on execution
4	ProductBasedReport export (supplementary aggregated data)	Location/category totals available; required Cognos Region × Product crosstab and Quantity output not independently verified
5	RetailSales Prompt Report	Prompt page visible; server error on run

2. Dataset Description

The available RetailPulse output contains sales information organised by location and product category. The ProductBasedReport export provides two location rows—New York and Long Island City—and an overall Total row. Product-category columns include Coffee, Flavours, Bakery, Coffee beans, Tea, Drinking Chocolate, Loose Tea, Packaged Chocolate and Branded. The overall reported sales total is $416,763.70. The available exports do not provide a reliably verified transaction-level row count; the ProductBasedReport contains two location records plus a Total row, with category totals shown in columns.

Available fields and structure

Area	Available information
Location	New York, Long Island City, Total
Product categories	Coffee, Flavours, Bakery, Coffee beans, Tea, Drinking Chocolate, Loose Tea, Packaged Chocolate, Branded
Measure	Visible in FirstSalesReport output; April–May 2024
Date information	Visible in FirstSalesReport output; dates span April–May 2024

3. Data Preparation

The exported ProductBasedReport was reviewed and used as the working dataset for the Power BI analysis. The data was imported into Power BI Desktop. Because the export is an aggregated report rather than a complete transaction-level table, the analysis was kept descriptive and focused on available location and category totals. No transaction-level trend, customer-level or causal claims were made.

Power BI preparation

The exported ProductBasedReport.xlsx was loaded into Power BI Desktop. The imported table was used to build comparative visuals. The DAX measure described later was created against the explicit Total row so that the report could display the overall sales value consistently.

4. IBM Cognos Analytics Work

Practical 1 — Generate a First RetailPulse Report

Practical 1 required a basic list report against the RetailPulse Sales package. The submitted FirstSalesReport output shows Sales, Product Category and Date columns. The report displays an overall sales value of $416,763.70 and product categories including Coffee, Packaged Chocolate, Drinking Chocolate, Coffee beans, Flavours, Bakery, Loose Tea, Tea and Branded. Dates shown in the output span April–May 2024.

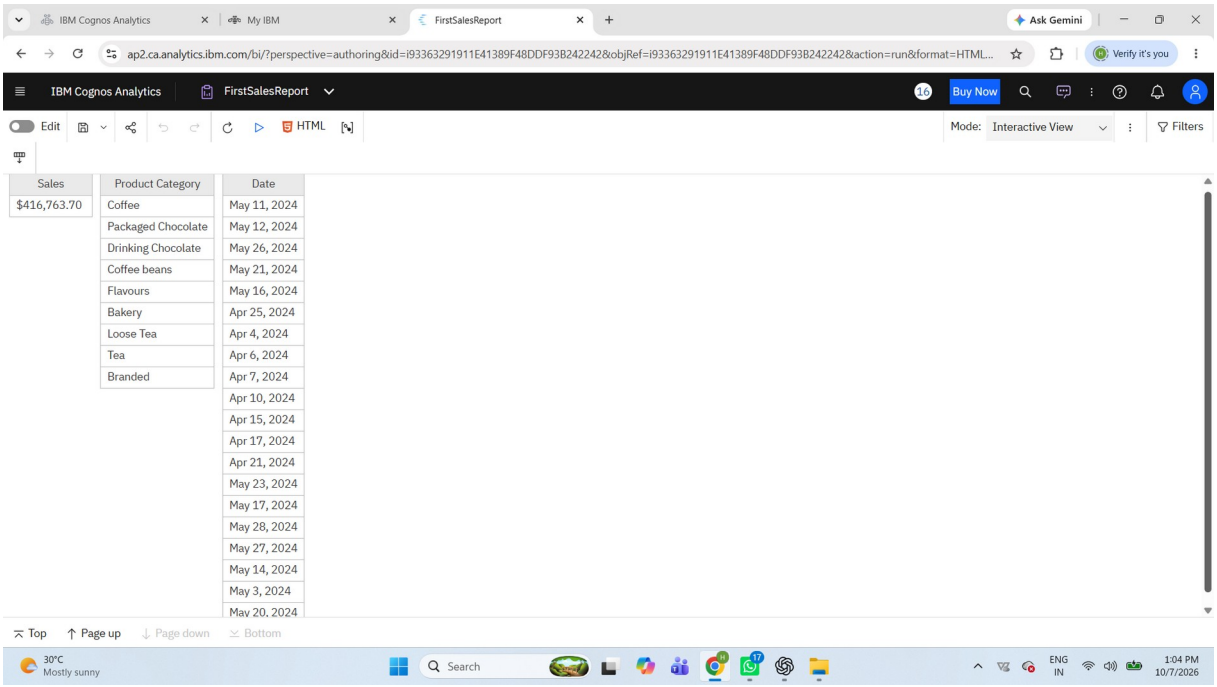


Figure 1: FirstSalesReport successfully displaying the RetailPulse sales output.

The practical instructions expected Region, Product Category and Sales Amount. The submitted output visibly contains Sales, Product Category and Date, so this report records the actual evidence and does not claim that a Region field was shown. The dates visible in the output span April–May 2024.

Practical 2 — Build a Grouped RetailPulse Sales List Report

Practical 2 required a grouped list report with Region and Store Name hierarchy, formatted Sales Amount, a total footer, and a page header. The report was created as SecondSalesReport. However, when execution was attempted, Cognos displayed the message 'The report could not run because a server error occurred.' Therefore, the expected grouped output and totals could not be verified from a successful run.

The report was intended to contain Region, Store Name, Product Category, Quantity and Sales Amount; Region and Store Name were to form a two-level grouping, with Sales Amount formatted as currency and a total footer added.

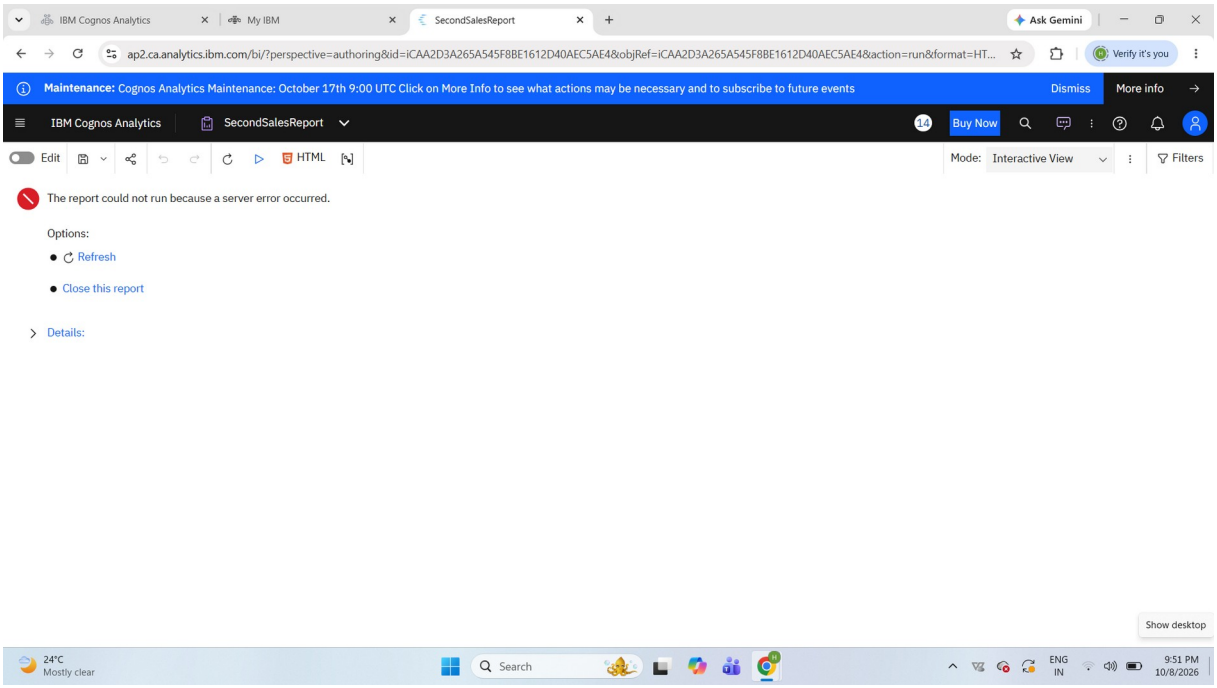


Figure 2: SecondSalesReport showing the Cognos server execution error.

Practical 3 — Filter RetailPulse Sales by Region and Date

Practical 3 required filtering the RetailPulse sales list to Region = North, applying an advanced Sales Date range filter, combining the conditions with AND, and converting the Region filter into a prompt. The report was created as ThirdSalesReport. Execution again returned a Cognos server error, so the final filtered row count and prompt-driven output could not be verified.

The report was intended to filter Region = North, apply a Sales Date range for the target quarter, combine the conditions using AND, and then convert the Region condition into a reusable prompt.

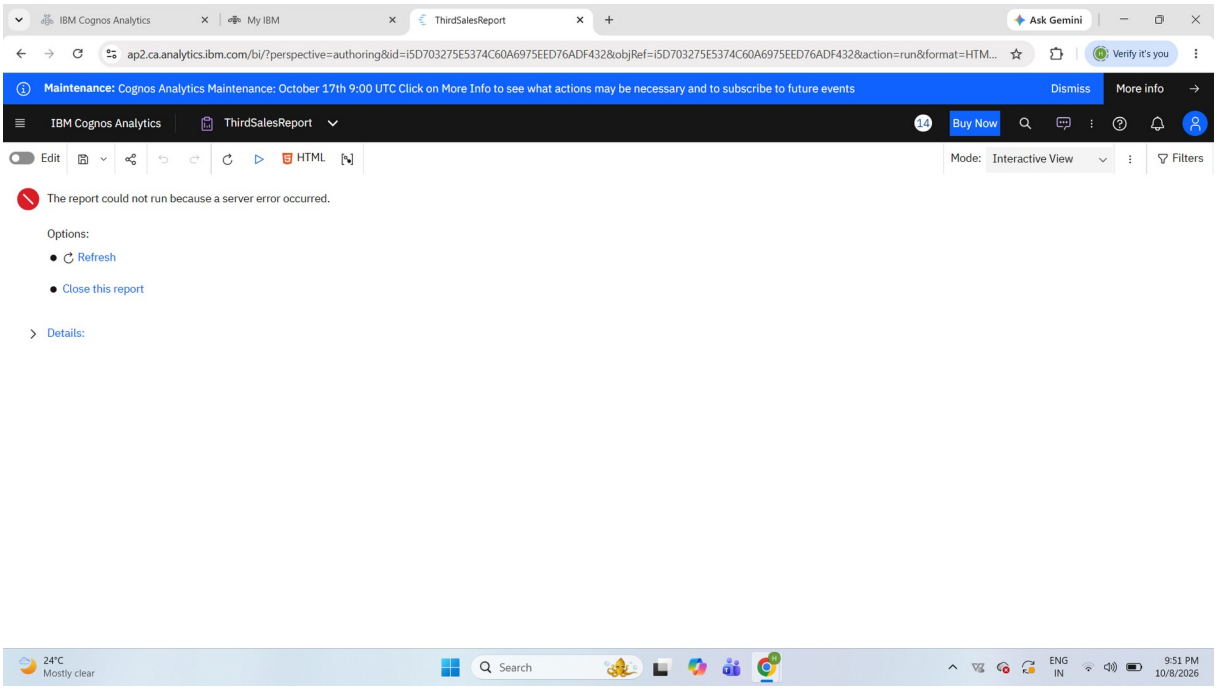


Figure 3: ThirdSalesReport showing the Cognos server execution error.

Practical 4 — Build a RetailPulse Region × Product Crosstab

Practical 4 required a Region × Product Category crosstab with Sales Amount and Quantity measures and summary totals. The available ProductBasedReport output provides the resulting regional and category sales totals. New York reports $277,712.32, Long Island City reports $139,051.38, and the overall total is $416,763.70. The category totals shown include Coffee $167,572.30, Flavours $2,772.80, Bakery $65,583.05, Coffee beans $14,031.15, Tea $111,470.80, Drinking Chocolate $44,359.00, Loose Tea $3,718.70, Packaged Chocolate $1,390.90 and Branded $5,865.00.

Observed totals in the supplementary ProductBasedReport export

Measure	New York	Long Island City	Total
Sales	$277,712.32	$139,051.38	$416,763.70
Coffee	$111,516.15	$56,056.15	$167,572.30
Flavours	$2,194.40	$578.40	$2,772.80
Bakery	$43,896.05	$21,687.00	$65,583.05
Coffee beans	$10,418.10	$3,613.05	$14,031.15
Tea	$73,387.15	$38,083.65	$111,470.80
Drinking Chocolate	$28,876.75	$15,482.25	$44,359.00
Loose Tea	$2,690.30	$1,028.40	$3,718.70
Packaged Chocolate	$1,030.42	$360.48	$1,390.90
Branded	$3,703.00	$2,162.00	$5,865.00

Figure 4: ProductBasedReport aggregated location and category sales export (used as supplementary data; not treated as verified proof of the required Region × Product Cognos crosstab).

Practical 5 — Build the RetailPulse Sales Prompt Report

Practical 5 required a prompt page containing Region, date-range and cascading Store controls linked to the crosstab report. The submitted prompt page visibly contains pRegion, Parameter2, pStore and pDate controls. The prompt page was displayed, but execution of the RetailSales Prompt Report then returned the same Cognos server-error message. Consequently, the final end-to-end prompt filtering could not be verified.

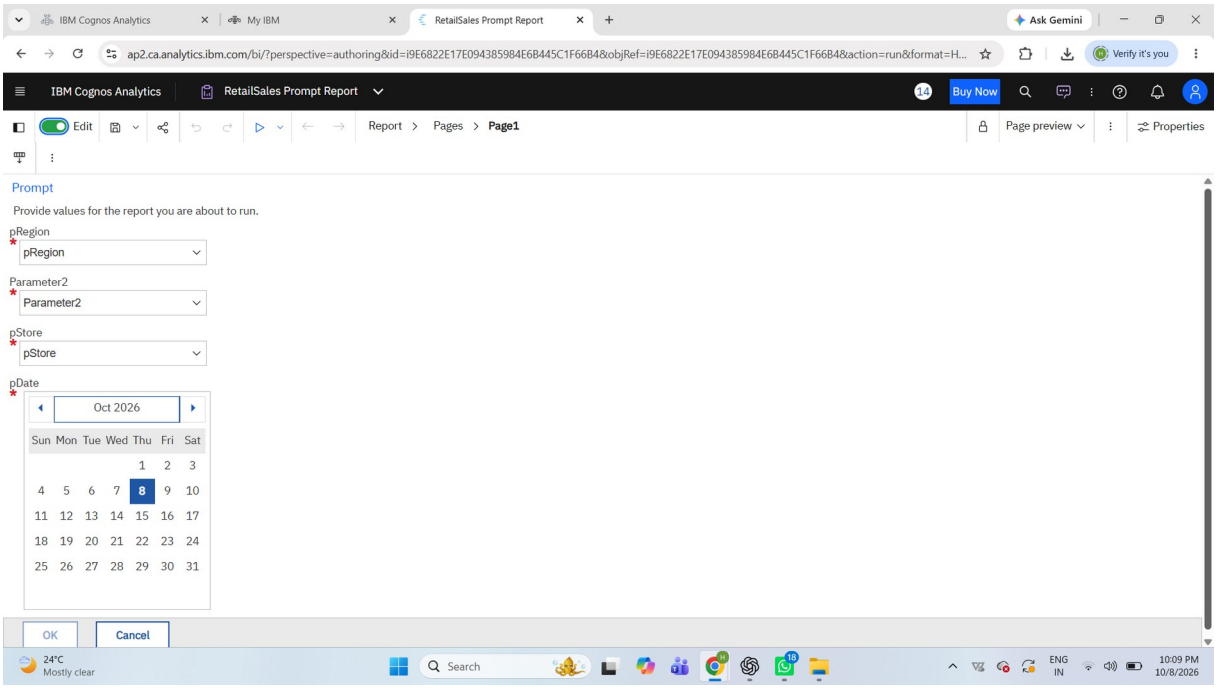


Figure 5: RetailSales Prompt Report showing the configured prompt controls.

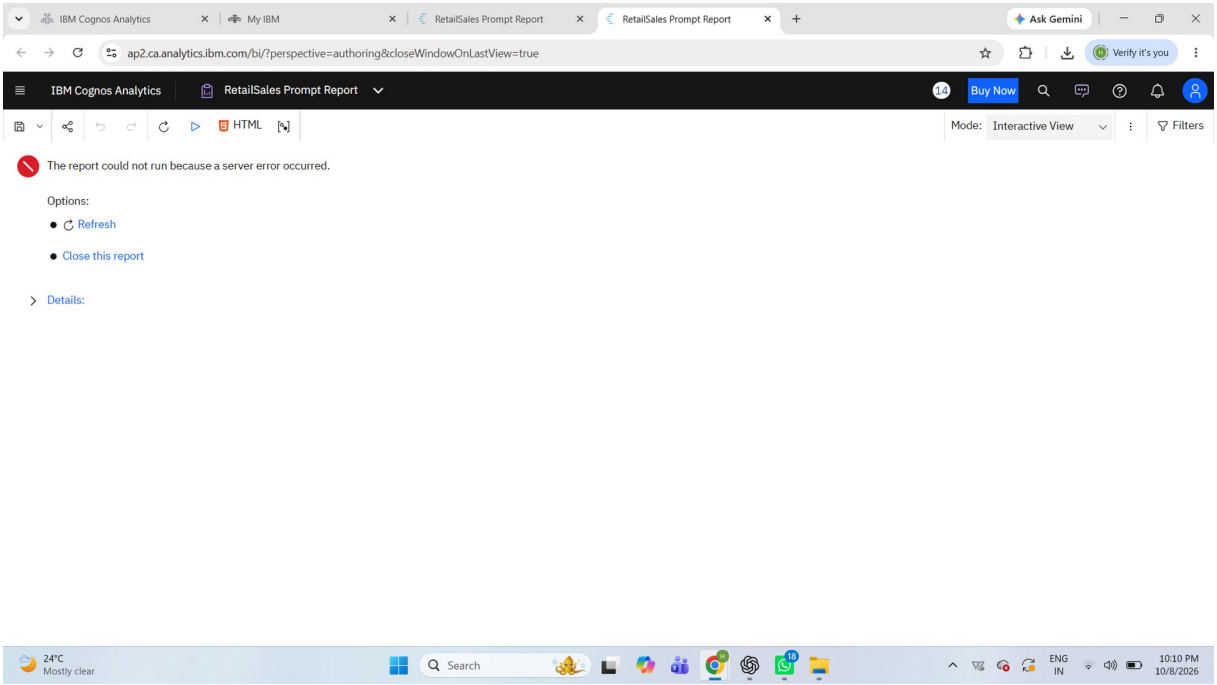


Figure 6: RetailSales Prompt Report showing the Cognos server execution error.

5. Power BI Analysis

The ProductBasedReport export was imported into Power BI Desktop. The imported fields were retained with the automatically generated Column1, Column2 and similar names where applicable. A DAX measure named Total Sales was created to retrieve the explicit Total row from the imported table. The report also contains comparative visuals for the available location/category values. The main overall sales value displayed is $416,763.70.

DAX measure

```
Total Sales =
CALCULATE(
    MAX('Page1'[Column11]),
    'Page1'[Column1] = "Total"
)
```

The measure filters the imported table to the row labelled Total and returns the corresponding overall sales value. This produced the reported overall value of $416,763.70.

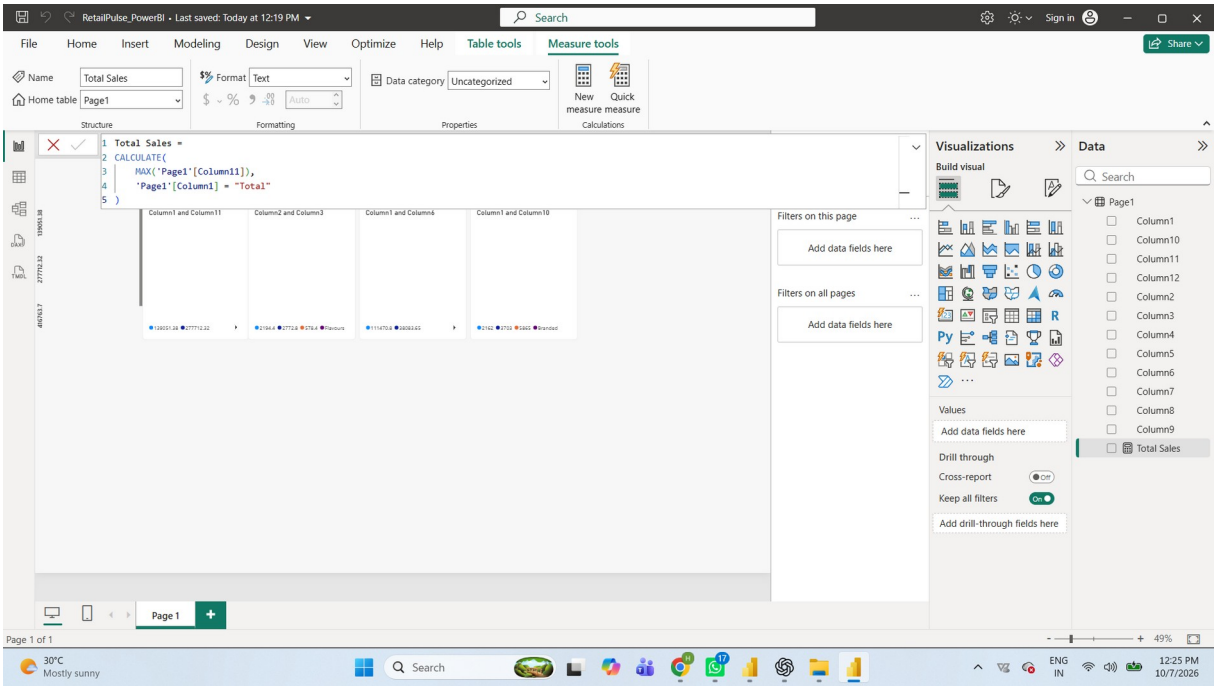


Figure 7: Power BI Desktop showing the Total Sales DAX measure and report visuals.

The saved editable Power BI file is RetailPulse_PowerBI_Final.pbix. A separate new Cognos report created for the Power BI workflow also returned the generic server-error message when run, even though it appeared correct in Edit mode. The Power BI visuals and measure documented here are based on the available ProductBasedReport export, not on a successful run of that separate Cognos report.

6. Findings and Insights

Finding	Value	Interpretation
Overall sales	$416,763.70	The available ProductBasedReport output reports total sales of $416,763.70.
New York	$277,712.32	New York contributes the larger share of the two reported locations.
Long Island City	$139,051.38	Long Island City contributes the remaining reported sales.
Location difference	$138,660.94	New York exceeds Long Island City by $138,660.94.
Coffee	$167,572.30	Coffee is a substantial reported category total in the crosstab output.
Flavours	$2,772.80	Flavours is a comparatively smaller reported category total.
Branded	$5,865.00	Branded contributes $5,865.00 to the reported total.

Location comparison

New York accounts for approximately 66.63% of the reported $416,763.70 total, while Long Island City accounts for approximately 33.37%. The difference between the two reported location totals is $138,660.94. These percentages are calculated from the exported totals.

7. Recommendations

1. Investigate the large sales gap between New York and Long Island City to identify differences in store performance, product mix, demand or operating conditions.

2. Maintain visibility on high-value categories such as Coffee while monitoring whether their contribution remains consistent in future periods.

3. Review lower-value categories such as Flavours and Branded to determine whether the difference is driven by demand, availability, pricing or assortment.

4. Extend the analysis with transaction-level, customer-level and time-series data so that future reports can evaluate trends, drivers and performance changes rather than only aggregated totals.

8. Limitations

- The available ProductBasedReport is aggregated and does not provide a complete transaction-level dataset for detailed trend or customer analysis.
- Practical 2, Practical 3 and Practical 5 could not be validated through successful execution because Cognos returned server errors. A separate new Cognos report created to support the Power BI work also returned a server error despite appearing correct in Edit mode. These generic messages do not identify the underlying cause, so this report records the observed issue without asserting with certainty that it was caused only by the server.
- The Practical 1 instructions specify Region, Product Category and Sales Amount, while the submitted FirstSalesReport output visibly contains Sales, Product Category and Date; the report therefore records the actual output rather than claiming an unseen Region field.
- The Power BI analysis is based on the available exported report structure, so conclusions should be treated as descriptive rather than causal.

9. Conclusion

The RetailPulse project demonstrates the use of IBM Cognos Analytics for structured reporting and Microsoft Power BI for visual descriptive analysis. The successful outputs establish an overall sales value of $416,763.70 and show a substantial difference between New York and Long Island City. The Power BI report provides an editable analytical layer through visuals and the Total Sales DAX measure.

The Cognos work also demonstrates an important reporting practice: technical execution problems should be documented rather than hidden. Practical 2, Practical 3 and Practical 5 were prepared but returned server errors during execution, so their unverified final results have not been presented as facts.

10. References

Student Report Submission Instructions, course-provided PDF.

RetailPulse Sales practical instructions supplied for Practicals 1–5.

Submitted IBM Cognos Analytics report outputs and screenshots.

Microsoft Power BI Desktop report: RetailPulse_PowerBI_Final.pbix.

11. Appendix — Submission File Checklist

File / folder	Status
Final descriptive report PDF	Prepared
Cognos practical evidence / exports	Prepared where available
Power BI PBIX	RetailPulse_PowerBI_Final.pbix
Screenshots	Practical 1–5 and Power BI evidence available
GitHub repository link	Add before final submission

GitHub repository: To be added before submission. The repository should contain the final report, supporting screenshots, Cognos exports/evidence, the PBIX file and any permitted supporting data.